

Intro to This Course, aka Why We Should Use Python

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Why learning Python?

- **Ultra mainstream and global leader in programming and data science!** Most used programming language in the world; extremely used in industry, but also increasingly adopted in academia. *Huge job market advantage as a data scientist, inside but especially outside academia!*
- **Huge community & ecosystem:** >600,000 packages on PyPI! (more than 30x compared to R);
- **Free & open-source**, like R... wherever you go, Python will be with you, for free!
- **Great for reproducibility:** like R, it integrates well with *GitHub, Quarto, Jupyter, RStudio, R* itself, and more; allows you to create reports, articles, books, web apps, experiments... A great tool for Open Science!
- **Strong programming language**, considered easier to read than R, and very widely used beyond statistical analysis (e.g., developing web apps, machine learning, deep learning & AI, programming experiments)



What you may expect to learn in this course

- Running fundamental operations and using core functions;
- Working with basic data types and structures (`list`, `dict`, `DataFrame`);
- Learning a bit of data manipulation for data science with `Pandas` and `NumPy`;
- Learning fundamental programming tools: loops, conditionals, custom functions;
- Brief overview of fancy stuff with machine learning, deep learning, natural language processing, experiment programming



Why should you use Python, later in your PhD?

I don't recommend using Python for doing the exact same things you could do with R. R is much more used in this PhD program and generally for statistical analysis in academia.

So why learning Python? Apart from its importance for “outside academia”, you could learn Python for your PhD for:

1. Advanced applications of **AI: pretrained models, Natural Language Processing** (e.g., sentiment analysis, text classification, embeddings);
2. **Computational efficiency** especially when handling **big data**;
3. **Behavioral experiment** (e.g., PsychoPy, OpenSesame are becoming widely used in cognitive science).



Python vs R (for us): Different Strengths

Task	R	Python
Data management for data science	 <code>base</code> , <code>dplyr</code> , <code>tidyverse</code> ; base R more specialized for statistics and data analysis	 <code>pandas</code> , <code>numpy</code> ; very flexible; more scalable on big data
Visualization	 <code>ggplot2</code> 	 <code>matplotlib</code> , <code>seaborn</code>
Statistical modeling	 <code>base</code> , <code>lme4</code> , <code>glmmTMB</code> , <code>mgcv</code> , <code>lavaan</code> , ...  	 many alternatives, but generally less flexible
Advanced machine learning	 <code>caret</code> , <code>mlr3</code>	 <code>scikit-learn</code> , industry standard
Deep learning & Natural Language Processing AI	 Limited, maybe <code>nnet</code> , <code>neuralnet</code>	 <code>transformers</code> , <code>keras</code> , <code>torch</code> ; Hugging Face platform  



Task	R	Python
Behavioral experiments	 Practically absent	 PsychoPy, OpenSesame

Python vs R (for us): increasingly integrated

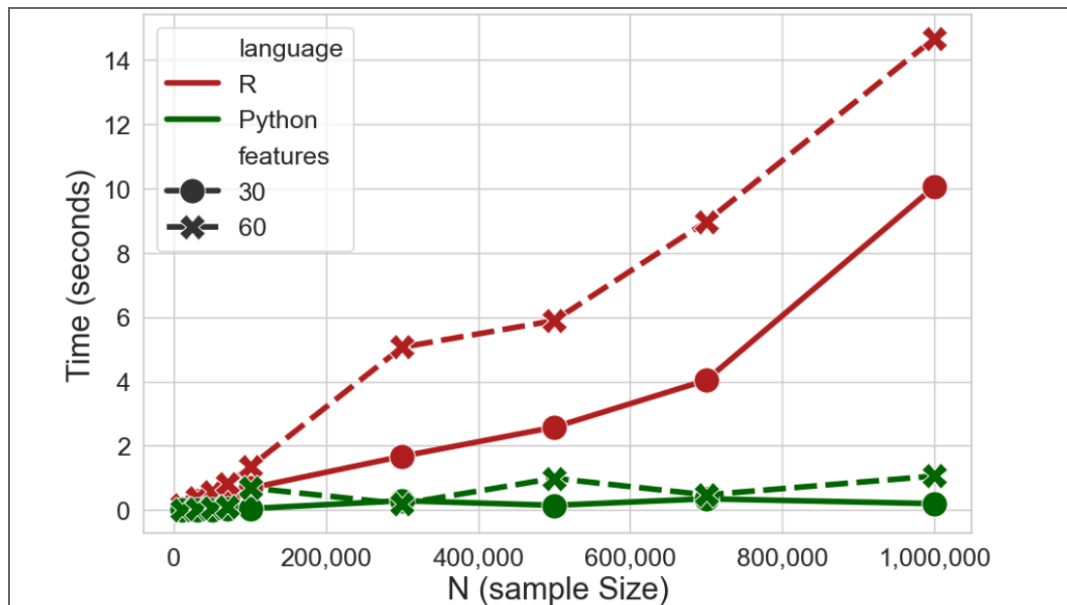
In fact, Python and R are becoming increasingly integrated, and often used together in research. For example:

- You can use Python in R using [reticulate](#) and R in Python using [rpy2](#);
- *RStudio* and *Quarto* can run Python script and chunks;
- *Jupyter* and *Colab* now allow R kernels;
- [shiny](#) and [ggplot](#) exist also for Python;
- [keras](#) and [tensorflow](#) can be used in R via [reticulate](#);
- [ellmer](#) allows you to use LLMs (including ChatGPT via API!) from inside R;
- [lme4](#) can be used inside Python via [pymer4](#);
- SEM can be run in Python using [semopy](#) (similar to [lavaan](#))



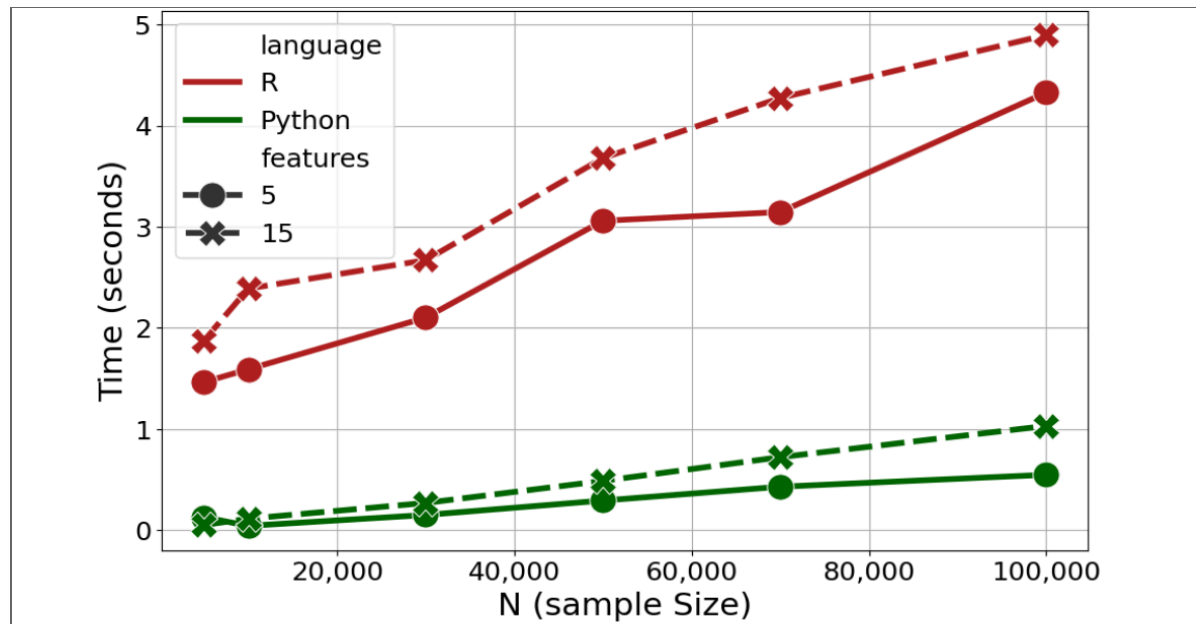
Python vs R Efficiency: Logistic Regression

► Some explanation (click to expand)



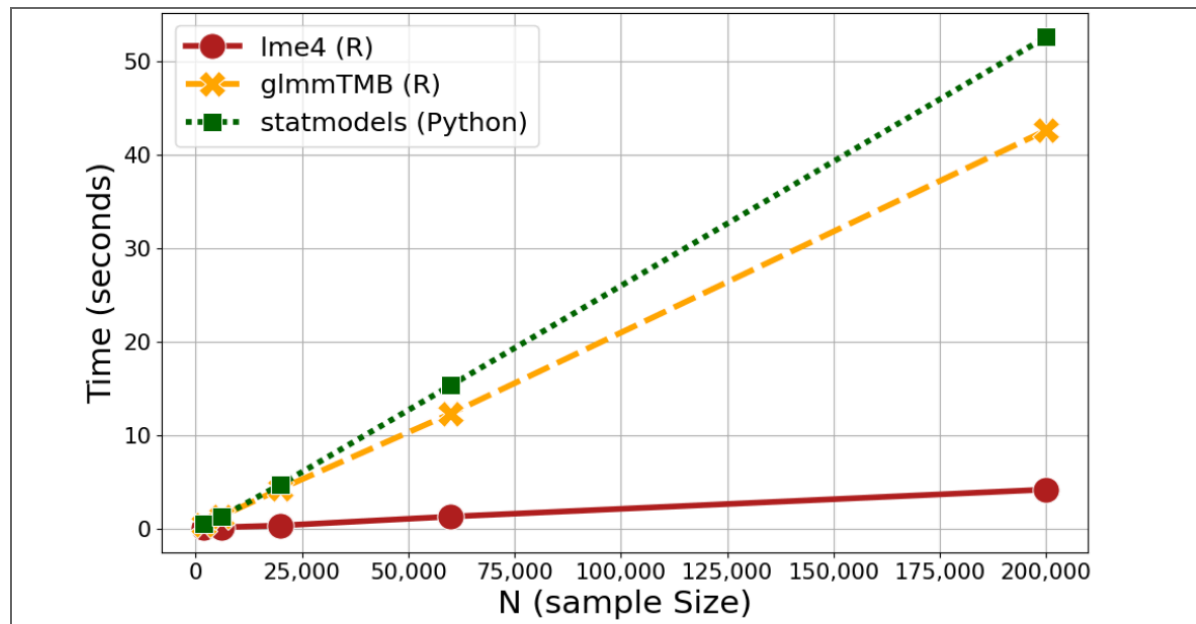
Python vs R Efficiency: GMM

► Some explanation (click to expand)



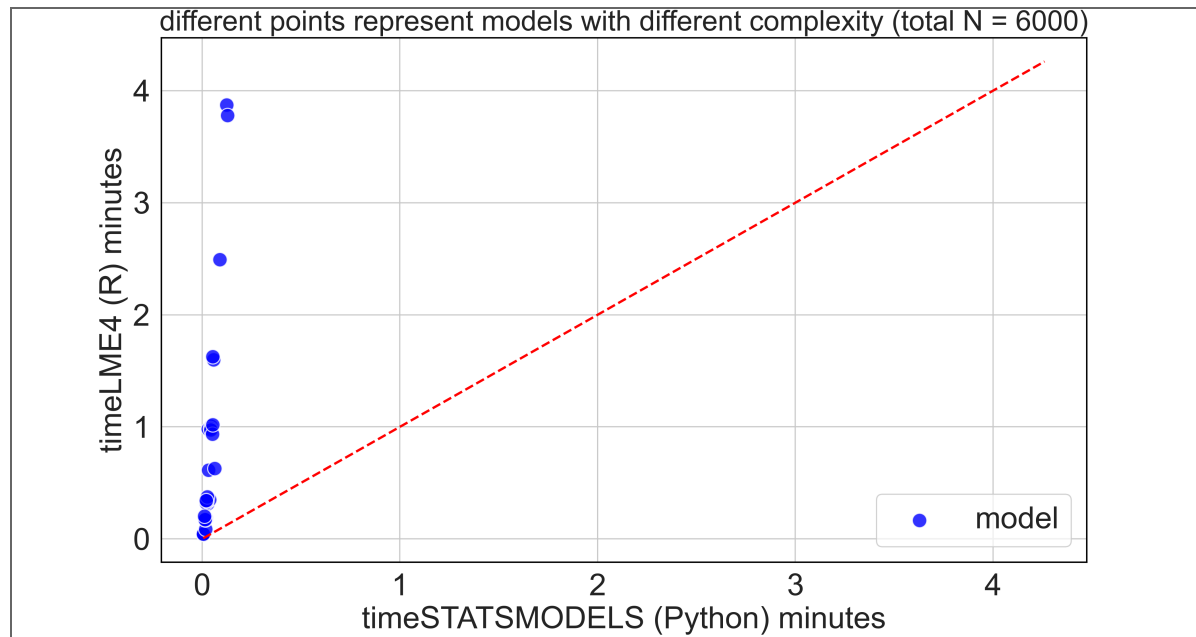
Python vs R Efficiency: Mixed-effects models

► Some explanation (click to expand)

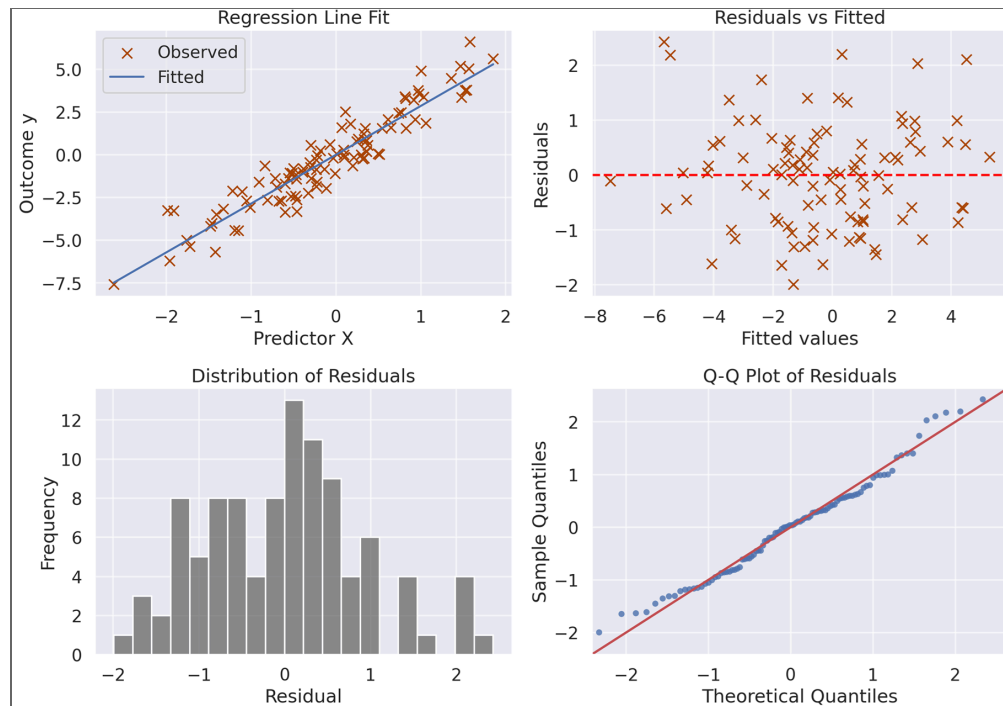


Python vs R Efficiency: Mixed-effects models

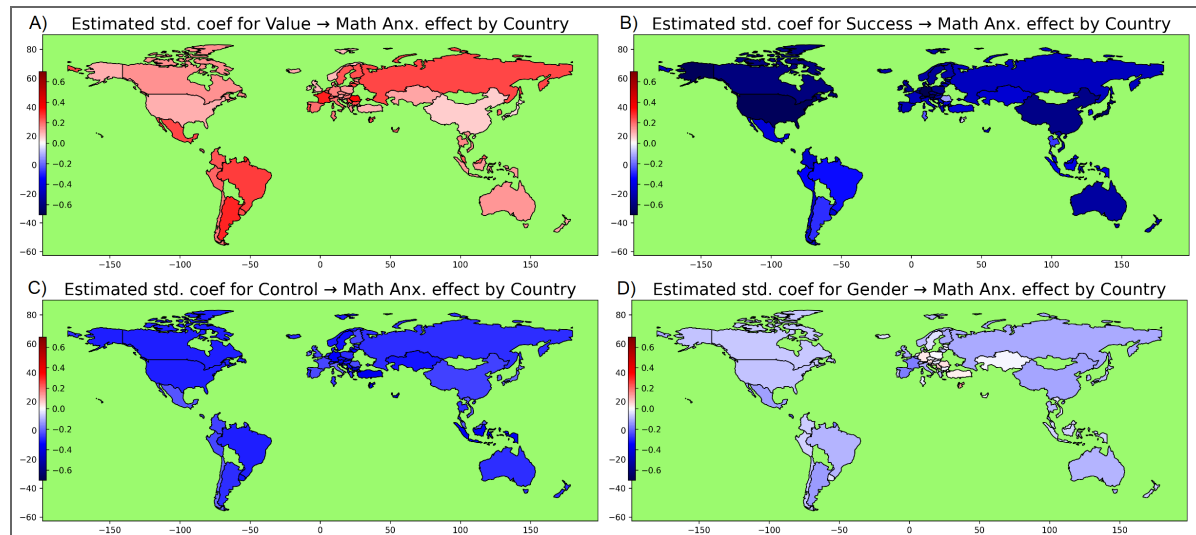
► Some explanation (click to expand)



Data visualization with **seaborn**

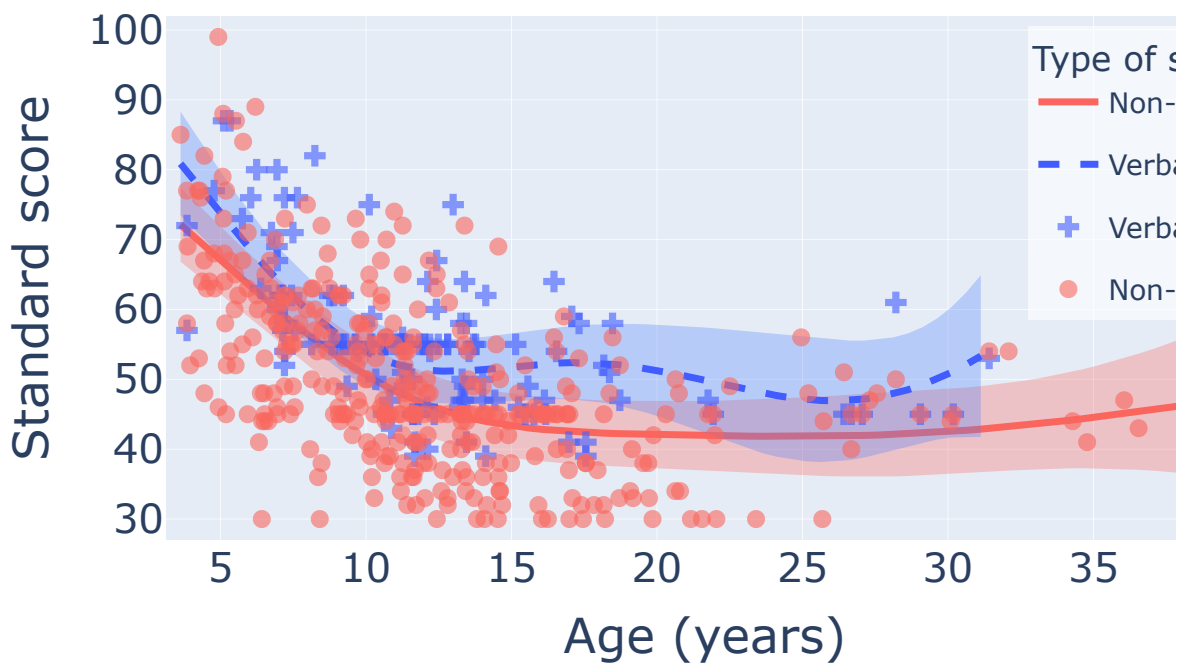


Data visualization with `matplotlib`



Interactive Data visualization with **plotly**

Change of Standard Scores by Area in I



Shiny app with Python

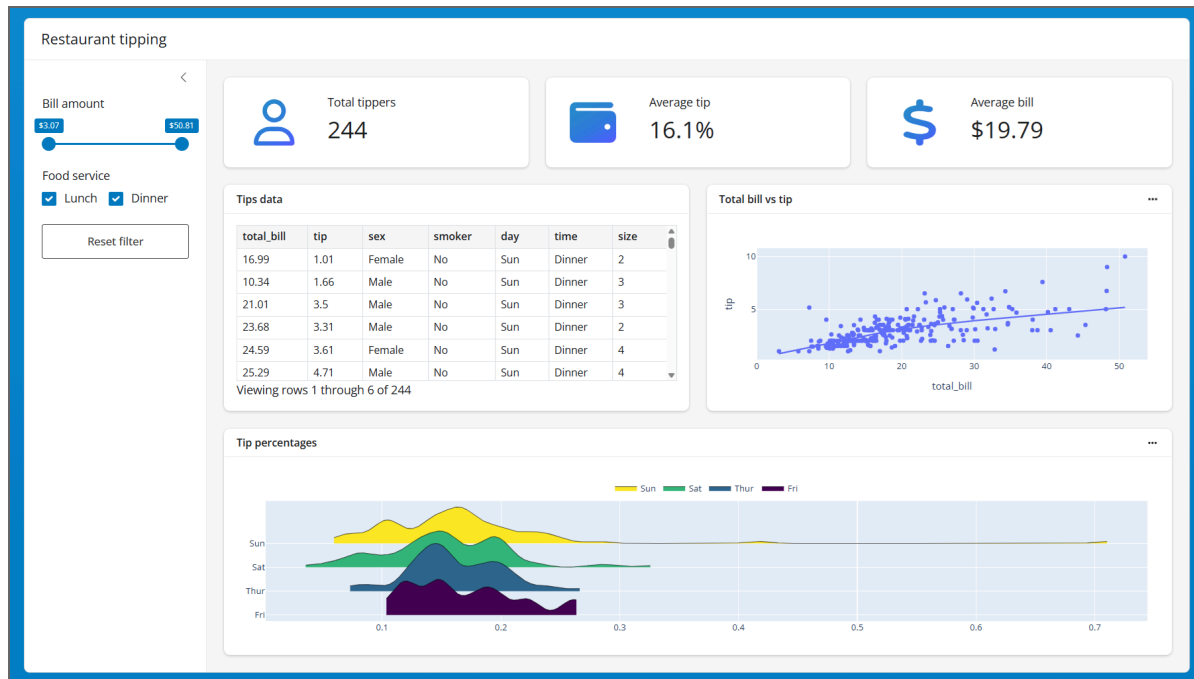


Image Classification with AI models via transformers



free model

used:

"microsoft/resnet-18"

score	label
0.2936	water snake
0.2809	Indian cobra, Naja naja
0.1630	ringneck snake, ring-necked snake, ring snake
0.1213	thunder snake, worm snake, Carphophis amoenus
0.0520	hognose snake, puff adder, sand viper

Text Classification with AI models via transformers

"That was absolutely terrifying. My hands were shaking, my heart pounded so loudly I could barely hear my own thoughts, and each creak of the floorboards seemed to scream danger. I stood frozen in place, staring at the door, every nerve in my body bracing for something awful to

free model

used:

"all-MiniLM-L12-v2"

target	cosineSimilarity
fear	0.4211
disgust	0.2082
surprise	0.1618
sadness	0.1528



joy
anger

0.1215
0.1098

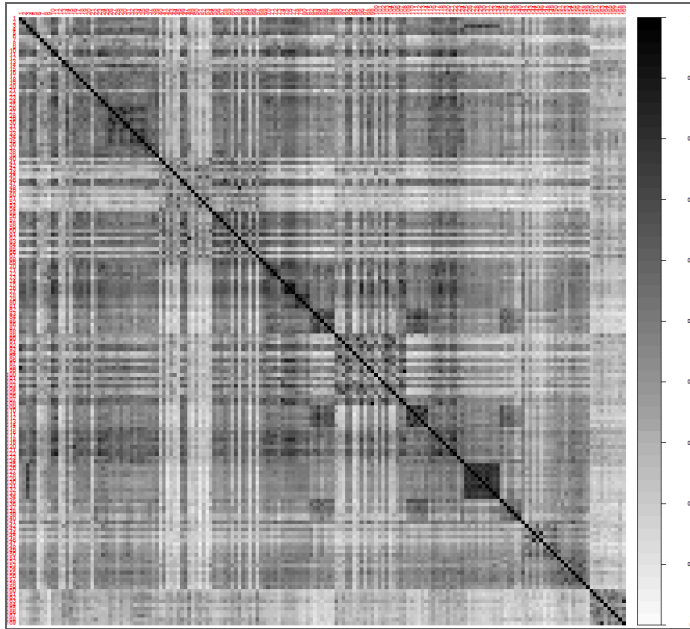


Text transcription from audio using transformers

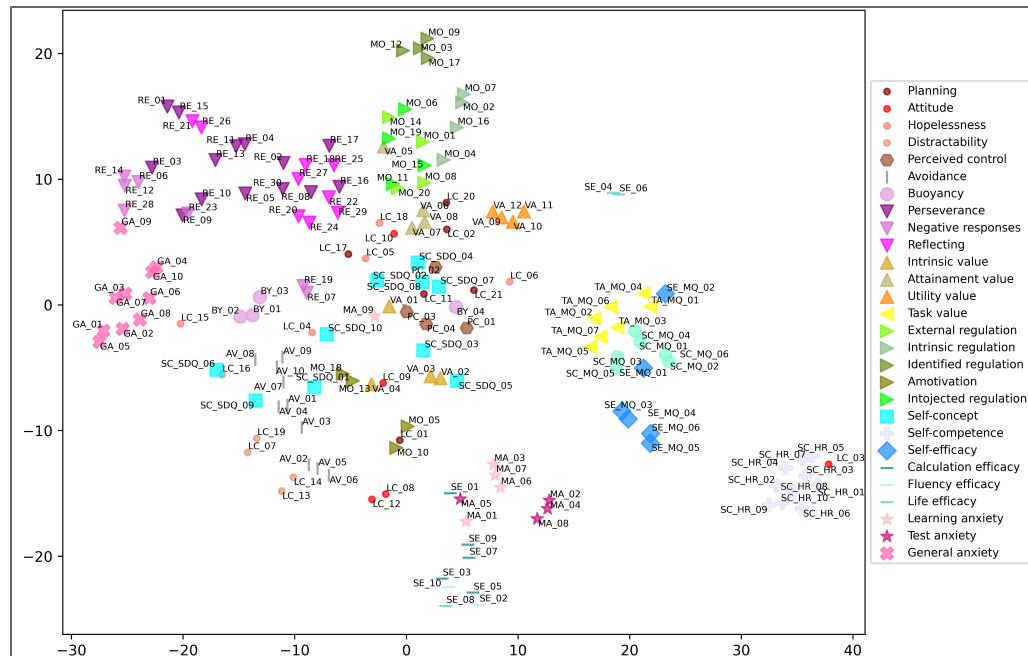
```
1 from transformers import pipeline
2 whisperpipe = pipeline(model="openai/w
3 whisperpipe("data/gelmanstats.mp3", re
```

Hi, my name is Andrew Gelman. I'm a professor of statistics and political science at Columbia. I've been teaching here since 1996. I'm always impressed that the students know more than I do about so many things. Just this morning, a graduate student came to me. He had been looking at data from ticketing, police moving violations. And there's this suspicion that the police have some sort of unofficial quota so that they're supposed to get a certain number of tickets by the end of the month. But you can look at the data and see whether you have more tickets at the very end of the month. He looked at something slightly different. He compared police that had had fewer tickets in the first half of the month to those who had

Working with semantic embeddings using LLMs (extracted with OpenAI's `text-embedding-ada-002`)



Working with semantic embeddings using LLMs (extracted with OpenAI's `text-embedding-ada-002`)



Embeddings + Clustering, An exercise we will see

